



AWS Well-Architected Tool

AWS Well-Architected Tool FTR_WAFR_FAILING - AWS Well-Architected Framework - FAILING Report

AWS Account ID: 156345700632

AWS Well-Architected Tool Report

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Table of contents

Milestone properties	4
Lens overview	6
Improvement plan	7
High risk	7
Medium risk	8
Lens details	10
Operational Excellence	10
Security	29
Reliability	46
Performance Efficiency	67
Cost Optimization	77
Sustainability	93

Milestone properties

Milestone name

FAILING

Date saved

Nov 3, 2025 5:44 PM UTC

Workload name

FTR_WAFR_FAILING

ARN

arn:aws:wellarchitected:us-east-1:156345700632:workload/
cda37f3de0301b19d8c8511d2cc32327

Description

Failing sample workload

Review owner

derekvi

Industry type

-

Industry

-

Environment

Production

AWS Regions

US East (N. Virginia)

Non-AWS regions

-

Account IDs

-

Architectural design

-

Application

-

Lens overview

Questions answered

56/57

Version

AWS Well-Architected Framework, 25th Feb 2025

Pillar	Questions answered
Operational Excellence	11/11
Security	10/11
Reliability	13/13
Performance Efficiency	5/5
Cost Optimization	11/11
Sustainability	6/6

Lens notes

-

Improvement plan

Improvement item summary

High risk: 2

Medium risk: 4

Pillar	High risk	Medium risk
Operational Excellence	1	2
Security	0	0
Reliability	1	2
Performance Efficiency	0	0
Cost Optimization	0	0
Sustainability	0	0

High risk

Operational Excellence
OPS 11.How do you evolve operations?

Security
No improvements identified

Reliability
REL 13.How do you plan for disaster recovery (DR)?

Performance Efficiency

No improvements identified

Cost Optimization

No improvements identified

Sustainability

No improvements identified

Medium risk

Operational Excellence

- [OPS 7.How do you know that you are ready to support a workload?](#)
- [OPS 10.How do you manage workload and operations events?](#)

Security

No improvements identified

Reliability

- [REL 9.How do you back up data?](#)
- [REL 12.How do you test reliability?](#)

Performance Efficiency
No improvements identified

Cost Optimization
No improvements identified

Sustainability
No improvements identified

Lens details

Operational Excellence

Questions answered

11/11

Question status

- ⊗ High risk: 1
- ⚠ Medium risk: 2
- ✓ No improvements identified: 8
- ⊖ Not Applicable: 0
- ⌚ Unanswered: 0

Pillar notes

-

1. How do you determine what your priorities are?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Evaluate compliance requirements
Out of Scope
- Evaluate threat landscape
Out of Scope
- Evaluate tradeoffs while managing benefits and risks
Out of Scope
- Evaluate external customer needs
Out of Scope
- Evaluate governance requirements
Out of Scope
- Evaluate internal customer needs
Out of Scope

Notes

OPS_1

Improvement plan

1. How do you determine what your priorities are?
No risk detected for this question. No action needed.

2. How do you structure your organization to support your business outcomes?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Mechanisms exist to manage responsibilities and ownership

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Operations activities have identified owners responsible for their performance

Out of Scope

- Responsibilities between teams are predefined or negotiated

Out of Scope

- Processes and procedures have identified owners

Out of Scope

- Resources have identified owners

Out of Scope

- Mechanisms exist to request additions, changes, and exceptions

Out of Scope

Notes

OPS_2: Provide answers for the following:

[WAFR-002] Have you reviewed the AWS Shared Responsibility Models for Security and Resiliency (Yes/No)? : Yes

2. How do you structure your organization to support your business outcomes?

[RES-005] Do you have a mechanism for communicating resiliency responsibility to customers (Yes/No)? : Yes

Improvement plan

No risk detected for this question. No action needed.

3. How does your organizational culture support your business outcomes?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Communications are timely, clear, and actionable
Out of Scope
- Provide executive sponsorship
Out of Scope
- Team members are empowered to take action when outcomes are at risk
Out of Scope
- Escalation is encouraged
Out of Scope
- Experimentation is encouraged
Out of Scope
- Team members are encouraged to maintain and grow their skill sets
Out of Scope
- Resource teams appropriately
Out of Scope

Notes

3. How does your organizational culture support your business outcomes?

OPS_3

Improvement plan

No risk detected for this question. No action needed.

4. How do you implement observability in your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement application telemetry
Out of Scope
- Implement user experience telemetry
Out of Scope
- Implement dependency telemetry
Out of Scope
- Implement distributed tracing
Out of Scope
- Identify key performance indicators
Out of Scope

Notes

OPS_4

Improvement plan

No risk detected for this question. No action needed.

5. How do you reduce defects, ease remediation, and improve flow into production?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Fully automate integration and deployment
Out of Scope
- Use build and deployment management systems
Out of Scope
- Implement practices to improve code quality
Out of Scope
- Use configuration management systems
Out of Scope
- Make frequent, small, reversible changes
Out of Scope
- Use multiple environments
Out of Scope
- Perform patch management
Out of Scope
- Share design standards

5. How do you reduce defects, ease remediation, and improve flow into production?

Out of Scope

- Test and validate changes

Out of Scope

- Use version control

Out of Scope

Notes

OPS_5

Improvement plan

No risk detected for this question. No action needed.

6. How do you mitigate deployment risks?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate testing and rollback
Out of Scope
- Employ safe deployment strategies
Out of Scope
- Plan for unsuccessful changes
Out of Scope
- Test deployments
Out of Scope

Notes

OPS_6

Improvement plan

No risk detected for this question. No action needed.

7. How do you know that you are ready to support a workload?

 Medium risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- Create support plans for production workloads
- None of these

Best Practices marked as Not Applicable

- Ensure a consistent review of operational readiness
Out of Scope
- Make informed decisions to deploy systems and changes
Out of Scope
- Ensure personnel capability
Out of Scope
- Use playbooks to investigate issues
Out of Scope
- Use runbooks to perform procedures
Out of Scope

Notes

OPS_7: Provide answers for the following:

[SUP-001] Is your support tier Business or higher on all production AWS Accounts (Yes/No)? : No

7. How do you know that you are ready to support a workload?

Improvement plan

- [Create support plans for production workloads](#)

[Ask an expert](#)

8. How do you utilize workload observability in your organization?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Analyze workload logs
Out of Scope
- Analyze workload metrics
Out of Scope
- Analyze workload traces
Out of Scope
- Create actionable alerts
Out of Scope
- Create dashboards
Out of Scope

Notes

OPS_8

Improvement plan

No risk detected for this question. No action needed.

9. How do you understand the health of your operations?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Communicate status and trends to ensure visibility into operation

Out of Scope

- Measure operations goals and KPIs with metrics

Out of Scope

- Review operations metrics and prioritize improvement

Out of Scope

Notes

OPS_9

Improvement plan

No risk detected for this question. No action needed.

10. How do you manage workload and operations events?

 Medium risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Define a customer communication plan for service-impacting events

Not selected choice(s)

- Communicate status through dashboards
- None of these

Best Practices marked as Not Applicable

- Automate responses to events
Out of Scope
- Define escalation paths
Out of Scope
- Use a process for event, incident, and problem management
Out of Scope
- Prioritize operational events based on business impact
Out of Scope
- Have a process per alert
Out of Scope

Notes

OPS_10

Improvement plan

- [Communicate status through dashboards](#)

10. How do you manage workload and operations events?

[Ask an expert](#)

11. How do you evolve operations?

⊗ High risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- Have a process for continuous improvement
- None of these

Best Practices marked as Not Applicable

- Allocate time to make improvements
Out of Scope
- Define drivers for improvement
Out of Scope
- Implement feedback loops
Out of Scope
- Perform knowledge management
Out of Scope
- Perform operations metrics reviews
Out of Scope
- Perform post-incident analysis
Out of Scope
- Document and share lessons learned
Out of Scope
- Validate insights

11. How do you evolve operations?

Out of Scope

Notes

OPS_11: Document how you implement the following best practice:

[WAFR-001] Do you do periodic architecture reviews (Yes/No)?: Yes

[WAFR-001] Have you done an architecture review in the last year (Yes/No)?: Yes

-- Note that if you are using this initial FTR as a starting point, you can create a milestone for the workload and then continue by completing all questions in the future after you have submitted for FTR.

Improvement plan

- [Have a process for continuous improvement](#)

[Ask an expert](#)

Security

Questions answered

10/11

Question status

-  High risk: 0
-  Medium risk: 0
-  No improvements identified: 10
-  Not Applicable: 0
-  Unanswered: 1

Pillar notes

-

1. How do you securely operate your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Secure account root user and properties
- Identify and validate control objectives

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate deployment of standard security controls
Out of Scope
- Evaluate and implement new security services and features regularly
Out of Scope
- Separate workloads using accounts
Out of Scope
- Reduce security management scope
Out of Scope
- Identify and prioritize risks using a threat model
Out of Scope
- Stay up to date with security threats and recommendations
Out of Scope

Notes

SEC_1: Provide answers for the following:

[ARC-001] Is the AWS Root Account only used for tasks listed in this link

1. How do you securely operate your workload?

<https://docs.aws.amazon.com/accounts/latest/reference/root-user-tasks.html> (Yes/No)?: Yes

[ARC-004] Have you removed/disabled Access keys for the Root User (Yes/No)?: Yes

[ARC-004] Have you included your CIS Benchmark report as an artifact (Yes/No)?: No

[RCVP-001] If you advertise the solution as meeting compliance standards, do you have a process to ensure compliance with the standards (Yes/No)? //remove if not applicable: Yes

Improvement plan

No risk detected for this question. No action needed.

2. How do you manage identities for people and machines?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Use strong sign-in mechanisms
- Use temporary credentials
- Store and use secrets securely
- Rely on a centralized identity provider
- Audit and rotate credentials periodically

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Employ user groups and attributes

Out of Scope

Notes

SEC_2: Provide answers for the following:

[IAM-001] Is MFA enabled for all human users to access AWS Accounts

-- this may be enabled in your IdP to federate access (Yes/No)? : Yes

[IAM-002] Are the credentials and access keys rotated at least every 90 days (Yes/No)? : Yes

[IAM-002] If static keys are used, do you maintain a list of where they are used (Yes/No)? //remove this question if not applicable: Yes

[IAM-002] Have you implemented monitoring of AWS CloudTrail logs to detect anomalous activity or other potential misuse (Yes/No)? : Yes

[IAM-002] Do you have a runbook or SOP for revoking credentials in

2. How do you manage identities for people and machines?

the event you detect a misuse (Yes/No)?: Yes

[IAM-003] Is a strong password policy enforced for users (Yes/No)?: Yes

[IAM-004] Does every individual user have their own unique credentials which are not shared with other users (Yes/No)?: Yes

[IAM-008] Was there an audit conducted for IAM users, roles and permissions in the last quarter (Yes/No)?: Yes

[IAM-009] Do you prevent credentials from being hard-coded in any application code (Yes/No)?: Yes

[IAM-010] Do you use secret management tools for encryption, access control, and access logging (Yes/No)?: Yes

[IAM-011] Do you encrypt end-user authentication credentials and passwords at rest (Yes/No)?: Yes

[IAM-012] Do you use IAM roles for all AWS API access from AWS compute services (e.g. Amazon EC2 Instance, AWS Lambda Functions, etc) (Yes/No)?: Yes

[IAM-012] If you are using static IAM credentials, are they stored securely (Yes/No)?: Yes

[CAA-006] Does your solutions architecture ensure provisioning read-only access to external IDs to customers (Yes/No)?: Yes

[CAA-005] Does your solution architecture ensure that all external IDs for external AWS Accounts are unique (Yes/No)?: Yes

[CAA-003] Does your solution architecture ensure any historical use of customer-provided IAM credentials are deprecated (Yes/No)?: Yes

Improvement plan

No risk detected for this question. No action needed.

3. How do you manage permissions for people and machines?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Grant least privilege access
- Define permission guardrails for your organization
- Manage access based on lifecycle
- Reduce permissions continuously
- Share resources securely with a third party
- Analyze public and cross account access

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Define access requirements
Out of Scope
- Establish emergency access process
Out of Scope
- Share resources securely within your organization
Out of Scope

Notes

SEC_3: Provide answers for the following:

[IAM-005] If third parties are given access to production workloads, do you only provide temporary instead of static credentials (Yes/No)? // remove if not applicable: Yes

[IAM-006] Is the principal of least privilege followed when granting

3. How do you manage permissions for people and machines?

access to users (Yes/No)?: Yes

[IAM-007] Is there a policy for when a user leaves the company or changes roles (Yes/No)?: Yes

[S3-001] Is there a process in place to periodically review access levels of S3 buckets (Yes/No)?: Yes

[CAA-001] Do you use cross-account roles to access customer AWS Accounts (Yes/No)?: Yes

[CAA-007] Do you provide guidance for customers on how to setup their AWS Account(s) for external access by your AWS Accounts (Yes/No)?: Yes

[CAA-002] Is external ID being used to access another customer's AWS Account (Yes/No)?: Yes

[CAA-004] Do you generate the values for external IDs to provide to customers (Yes/No)?: Yes

Improvement plan

No risk detected for this question. No action needed.

4. How do you detect and investigate security events?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Configure service and application logging
Out of Scope
- Capture logs, findings, and metrics in standardized locations
Out of Scope
- Initiate remediation for non-compliant resources
Out of Scope
- Correlate and enrich security events
Out of Scope

Notes

SEC_4

Improvement plan

No risk detected for this question. No action needed.

5. How do you protect your network resources?

 Unanswered

Selected choice(s)

-

Not selected choice(s)

- Create network layers
- Control traffic within your network layers
- Implement inspection-based protection
- Automate network protection
- None of these

Best Practices marked as Not Applicable

-

Notes

SEC_5: Provide answers for the following:

[NETSEC-001] Does the default security group restrict all traffic (Yes/No)?: No

[NETSEC-001] Are all security groups configured to not all ingress from 0.0.0.0/0 to port 22 or 3389 (Yes/No)?: No

[NETSEC-002] Are all of the resources hosted in private subnets (Yes/No)?: No

Improvement plan

Answer the question to view the improvement plan.

6. How do you protect your compute resources?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Perform vulnerability management
- Automate compute protection

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Provision compute from hardened images
Out of Scope
- Reduce manual management and interactive access
Out of Scope
- Validate software integrity
Out of Scope

Notes

SEC_6: Provide answers for the following:

[SECOPS-001] Is there a process to scan software and patch vulnerabilities on a scheduled basis (Yes/No)? : Yes

Improvement plan

No risk detected for this question. No action needed.

7. How do you classify your data?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Understand your data classification scheme
- Apply data protection controls based on data sensitivity

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate identification and classification
Out of Scope
- Define scalable data lifecycle management
Out of Scope

Notes

SEC_7: Provide answers for the following:

[SDAT-001] Is there a process to perform reviews to classify your data (Yes/No)? Yes

Improvement plan

No risk detected for this question. No action needed.

8. How do you protect your data at rest?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Enforce encryption at rest

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Enforce access control
Out of Scope
- Automate data at rest protection
Out of Scope
- Implement secure key management
Out of Scope

Notes

SEC_8

Improvement plan

No risk detected for this question. No action needed.

9. How do you protect your data in transit?

✔ No improvements identified

Selected choice(s)

- Implement secure key and certificate management
- Enforce encryption in transit
- Authenticate network communications

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

-

Notes

SEC_9

Improvement plan

No risk detected for this question. No action needed.

10. How do you anticipate, respond to, and recover from incidents?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Identify key personnel and external resources
- Develop incident management plans

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Establish a framework for learning from incidents
Out of Scope
- Develop and test security incident response playbooks
Out of Scope
- Pre-deploy tools
Out of Scope
- Pre-provision access
Out of Scope
- Prepare forensic capabilities
Out of Scope
- Run simulations
Out of Scope

Notes

SEC_10: Provide answers for the following:

[ARC-005] Is there a documented process to respond, mitigate, and

10. How do you anticipate, respond to, and recover from incidents?

recover from security incidents (Yes/No)?: Yes

[ACOM-001] Are AWS alternate contacts configured (Yes/No)?: Yes

Improvement plan

No risk detected for this question. No action needed.

11. How do you incorporate and validate the security properties of applications throughout the design, development, and deployment lifecycle?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate testing throughout the development and release lifecycle
Out of Scope
- Build a program that embeds security ownership in workload teams
Out of Scope
- Centralize services for packages and dependencies
Out of Scope
- Deploy software programmatically
Out of Scope
- Conduct code reviews
Out of Scope
- Perform regular penetration testing
Out of Scope
- Regularly assess security properties of the pipelines
Out of Scope

11. How do you incorporate and validate the security properties of applications throughout the design, development, and deployment lifecycle?

- Train for application security

Out of Scope

Notes

SEC_11

Improvement plan

No risk detected for this question. No action needed.

Reliability

Questions answered

13/13

Question status

-  High risk: 1
-  Medium risk: 2
-  No improvements identified: 10
-  Not Applicable: 0
-  Unanswered: 0

Pillar notes

-

1. How do you manage service quotas and constraints?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Aware of service quotas and constraints

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate quota management
Out of Scope
- Accommodate fixed service quotas and constraints through architecture
Out of Scope
- Manage service quotas across accounts and Regions
Out of Scope
- Monitor and manage quotas
Out of Scope
- Ensure that a sufficient gap exists between the current quotas and the maximum usage to accommodate failover
Out of Scope

Notes

REL_1

Improvement plan

No risk detected for this question. No action needed.

2. How do you plan your network topology?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Provision redundant connectivity between private networks in the cloud and on-premises environments

Out of Scope

- Use highly available network connectivity for your workload public endpoints

Out of Scope

- Ensure IP subnet allocation accounts for expansion and availability

Out of Scope

- Enforce non-overlapping private IP address ranges in all private address spaces where they are connected

Out of Scope

- Prefer hub-and-spoke topologies over many-to-many mesh

Out of Scope

Notes

REL_2

2. How do you plan your network topology?

Improvement plan

No risk detected for this question. No action needed.

3. How do you design your workload service architecture?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

- Build services focused on specific business domains and functionality
- Provide service contracts per API

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Choose how to segment your workload

Out of Scope

Notes

REL_3: Provide answers for the following:

[RES-006] Does the solution have SLAs defined for availability and uptime (Yes/No)?: Yes

[RES-006] Does your architecture support the SLA (Yes/No)?: Yes

Improvement plan

No risk detected for this question. No action needed.

4. How do you design interactions in a distributed system to prevent failures?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Do constant work
Out of Scope
- Make mutating operations idempotent
Out of Scope
- Identify the kind of distributed systems you depend on
Out of Scope
- Implement loosely coupled dependencies
Out of Scope

Notes

REL_4

Improvement plan

No risk detected for this question. No action needed.

5. How do you design interactions in a distributed system to mitigate or withstand failures?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Set client timeouts
Out of Scope
- Implement emergency levers
Out of Scope
- Fail fast and limit queues
Out of Scope
- Implement graceful degradation to transform applicable hard dependencies into soft dependencies
Out of Scope
- Control and limit retry calls
Out of Scope
- Make systems stateless where possible
Out of Scope
- Throttle requests
Out of Scope

5. How do you design interactions in a distributed system to mitigate or withstand failures?

Notes

REL_5

Improvement plan

No risk detected for this question. No action needed.

6. How do you monitor workload resources?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate responses (Real-time processing and alarming)
Out of Scope
- Monitor end-to-end tracing of requests through your system
Out of Scope
- Monitor all components for the workload (Generation)
Out of Scope
- Define and calculate metrics (Aggregation)
Out of Scope
- Send notifications (Real-time processing and alarming)
Out of Scope
- Regularly review monitoring scope and metrics
Out of Scope
- Analyze logs
Out of Scope

Notes

REL_6

6. How do you monitor workload resources?

Improvement plan

No risk detected for this question. No action needed.

7. How do you design your workload to adapt to changes in demand?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Use automation when obtaining or scaling resources
Out of Scope
- Load test your workload
Out of Scope
- Obtain resources upon detection that more resources are needed for a workload
Out of Scope
- Obtain resources upon detection of impairment to a workload
Out of Scope

Notes

REL_7

Improvement plan

No risk detected for this question. No action needed.

8. How do you implement change?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Deploy changes with automation
Out of Scope
- Integrate functional testing as part of your deployment
Out of Scope
- Deploy using immutable infrastructure
Out of Scope
- Use runbooks for standard activities such as deployment
Out of Scope
- Integrate resiliency testing as part of your deployment
Out of Scope

Notes

REL_8

Improvement plan

No risk detected for this question. No action needed.

9. How do you back up data?

 Medium risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- Perform data backup automatically
- Perform periodic recovery of the data to verify backup integrity and processes
- None of these

Best Practices marked as Not Applicable

- Identify and back up all data that needs to be backed up, or reproduce the data from sources

Out of Scope

- Secure and encrypt backups

Out of Scope

Notes

REL_9: Provide answers for the following:

[BAR-001] Is all critical production data backed up automatically (Yes/No)?: No

[BAR-002] Do you perform periodic recovery of the data to verify backup integrity and processes (Yes/No)?: No

[BAR-002] Provide the date for the last time your disaster recovery plan was tested:

[BAR-002] Do you test data recovery after making significant changes

9. How do you back up data?

to the solution (Yes/No)?: No

Improvement plan

- Perform data backup automatically
- Perform periodic recovery of the data to verify backup integrity and processes

[Ask an expert](#)

10. How do you use fault isolation to protect your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Deploy the workload to multiple locations
Out of Scope
- Automate recovery for components constrained to a single location
Out of Scope
- Use bulkhead architectures to limit scope of impact
Out of Scope

Notes

REL_10

Improvement plan

No risk detected for this question. No action needed.

11. How do you design your workload to withstand component failures?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Automate healing on all layers
Out of Scope
- Rely on the data plane and not the control plane during recovery
Out of Scope
- Fail over to healthy resources
Out of Scope
- Monitor all components of the workload to detect failures
Out of Scope
- Send notifications when events impact availability
Out of Scope
- Architect your product to meet availability targets and uptime service level agreements (SLAs)
Out of Scope
- Use static stability to prevent bimodal behavior
Out of Scope

11. How do you design your workload to withstand component failures?

Notes

REL_11

Improvement plan

No risk detected for this question. No action needed.

12. How do you test reliability?

 Medium risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- Test resiliency using chaos engineering
- Conduct game days regularly
- None of these

Best Practices marked as Not Applicable

- Use playbooks to investigate failures
Out of Scope
- Perform post-incident analysis
Out of Scope
- Test scalability and performance requirements
Out of Scope

Notes

REL_12: Provide answers for the following:

[RES-004] What was the date of your last resiliency test?: TBD

[RES-004] What is your effective RPO?: TBD

[RES-004] What is your effective RTO?: TBD

Improvement plan

- [Test resiliency using chaos engineering](#)
- [Conduct game days regularly](#)

12. How do you test reliability?
Ask an expert

13. How do you plan for disaster recovery (DR)?

⊗ High risk

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- Define recovery objectives for downtime and data loss
- Test disaster recovery implementation to validate the implementation
- None of these

Best Practices marked as Not Applicable

- Automate recovery
Out of Scope
- Manage configuration drift at the DR site or Region
Out of Scope
- Use defined recovery strategies to meet the recovery objectives
Out of Scope

Notes

REL_13: Provide answers for the following:

[RES-001] What is the defined RPO time for your solution?: TBD

[RES-002] What is the defined RTO time for your solution?: TBD

Improvement plan

- Define recovery objectives for downtime and data loss
- Test disaster recovery implementation to validate the implementation

13. How do you plan for disaster recovery (DR)?

[Ask an expert](#)

Performance Efficiency

Questions answered

5/5

Question status

- ⊗ High risk: 0
- ⚠ Medium risk: 0
- ✓ No improvements identified: 5
- ⊖ Not Applicable: 0
- 🕒 Unanswered: 0

Pillar notes

-

1. How do you select the appropriate cloud resources and architecture patterns for your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Evaluate how trade-offs impact customers and architecture efficiency
Out of Scope
- Factor cost into architectural decisions
Out of Scope
- Use guidance from your cloud provider or an appropriate partner to learn about architecture patterns and best practices
Out of Scope
- Learn about and understand available cloud services and features
Out of Scope
- Use benchmarking to drive architectural decisions
Out of Scope
- Use a data-driven approach for architectural choices
Out of Scope
- Use policies and reference architectures
Out of Scope

1. How do you select the appropriate cloud resources and architecture patterns for your workload?

Notes

PERF_1

Improvement plan

No risk detected for this question. No action needed.

2. How do you select and use compute resources in your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Collect compute-related metrics
Out of Scope
- Use optimized hardware-based compute accelerators
Out of Scope
- Configure and right-size compute resources
Out of Scope
- Scale your compute resources dynamically
Out of Scope
- Select the best compute options for your workload
Out of Scope
- Understand the available compute configuration and features
Out of Scope

Notes

PERF_2

Improvement plan

2. How do you select and use compute resources in your workload?

No risk detected for this question. No action needed.

3. How do you store, manage, and access data in your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement data access patterns that utilize caching
Out of Scope
- Collect and record data store performance metrics
Out of Scope
- Evaluate available configuration options for data store
Out of Scope
- Implement strategies to improve query performance in data store
Out of Scope
- Use purpose-built data store that best support your data access and storage requirements
Out of Scope

Notes

PERF_3

Improvement plan

No risk detected for this question. No action needed.

4. How do you select and configure networking resources in your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Choose appropriate dedicated connectivity or VPN for your workload
Out of Scope
- Choose network protocols to improve performance
Out of Scope
- Choose your workload's location based on network requirements
Out of Scope
- Evaluate available networking features
Out of Scope
- Use load balancing to distribute traffic across multiple resources
Out of Scope
- Optimize network configuration based on metrics
Out of Scope
- Understand how networking impacts performance
Out of Scope

Notes

4. How do you select and configure networking resources in your workload?

PERF_4

Improvement plan

No risk detected for this question. No action needed.

5. What process do you use to support more performance efficiency for your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Use automation to proactively remediate performance-related issues
Out of Scope
- Establish key performance indicators (KPIs) to measure workload health and performance
Out of Scope
- Keep your workload and services up-to-date
Out of Scope
- Load test your workload
Out of Scope
- Review metrics at regular intervals
Out of Scope
- Use monitoring solutions to understand the areas where performance is most critical
Out of Scope
- Define a process to improve workload performance

5. What process do you use to support more performance efficiency for your workload?

Out of Scope

Notes

PERF_5

Improvement plan

No risk detected for this question. No action needed.

Cost Optimization

Questions answered

11/11

Question status

-  High risk: 0
-  Medium risk: 0
-  No improvements identified: 11
-  Not Applicable: 0
-  Unanswered: 0

Pillar notes

-

1. How do you implement cloud financial management?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Establish cloud budgets and forecasts
Out of Scope
- Implement cost awareness in your organizational processes
Out of Scope
- Create a cost-aware culture
Out of Scope
- Establish ownership of cost optimization
Out of Scope
- Establish a partnership between finance and technology
Out of Scope
- Monitor cost proactively
Out of Scope
- Quantify business value from cost optimization
Out of Scope
- Keep up-to-date with new service releases
Out of Scope

1. How do you implement cloud financial management?

- Report and notify on cost optimization

Out of Scope

Notes

COST_1

Improvement plan

No risk detected for this question. No action needed.

2. How do you govern usage?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement an account structure
Out of Scope
- Implement cost controls
Out of Scope
- Implement goals and targets
Out of Scope
- Implement groups and role
Out of Scope
- Develop policies based on your organization requirements
Out of Scope
- Track project lifecycle
Out of Scope

Notes

COST_2

Improvement plan

2. How do you govern usage?
No risk detected for this question. No action needed.

3. How do you monitor your cost and usage?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Allocate costs based on workload metrics
Out of Scope
- Configure billing and cost management tools
Out of Scope
- Identify cost attribution categories
Out of Scope
- Establish organization metrics
Out of Scope
- Configure detailed information sources
Out of Scope
- Add organization information to cost and usage
Out of Scope

Notes

COST_3

Improvement plan

3. How do you monitor your cost and usage?
No risk detected for this question. No action needed.

4. How do you decommission resources?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Enforce data retention policies
Out of Scope
- Decommission resources automatically
Out of Scope
- Decommission resources
Out of Scope
- Implement a decommissioning process
Out of Scope
- Track resources over their life time
Out of Scope

Notes

COST_4

Improvement plan

No risk detected for this question. No action needed.

5. How do you evaluate cost when you select services?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Analyze all components of this workload
Out of Scope
- Perform cost analysis for different usage over time
Out of Scope
- Select software with cost effective licensing
Out of Scope
- Identify organization requirements for cost
Out of Scope
- Select components of this workload to optimize cost in line with organization priorities
Out of Scope
- Perform a thorough analysis of each component
Out of Scope

Notes

COST_5

5. How do you evaluate cost when you select services?

Improvement plan

No risk detected for this question. No action needed.

6. How do you meet cost targets when you select resource type, size and number?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Perform cost modeling
Out of Scope
- Select resource type, size, and number based on data
Out of Scope
- Select resource type, size, and number automatically based on metrics
Out of Scope
- Consider using shared resources
Out of Scope

Notes

COST_6

Improvement plan

No risk detected for this question. No action needed.

7. How do you use pricing models to reduce cost?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Perform pricing model analysis
Out of Scope
- Implement pricing models for all components of this workload
Out of Scope
- Perform pricing model analysis at the management account level
Out of Scope
- Choose Regions based on cost
Out of Scope
- Select third-party agreements with cost-efficient terms
Out of Scope

Notes

COST_7

Improvement plan

No risk detected for this question. No action needed.

8. How do you plan for data transfer charges?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement services to reduce data transfer costs
Out of Scope
- Perform data transfer modeling
Out of Scope
- Select components to optimize data transfer cost
Out of Scope

Notes

COST_8

Improvement plan

No risk detected for this question. No action needed.

9. How do you manage demand, and supply resources?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement a buffer or throttle to manage demand

Out of Scope

- Perform an analysis on the workload demand

Out of Scope

- Supply resources dynamically

Out of Scope

Notes

COST_9

Improvement plan

No risk detected for this question. No action needed.

10. How do you evaluate new services?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Develop a workload review process
Out of Scope
- Review and analyze this workload regularly
Out of Scope

Notes

COST_10

Improvement plan

No risk detected for this question. No action needed.

11. How do you evaluate the cost of effort?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Perform automation for operations

Out of Scope

Notes

COST_11

Improvement plan

No risk detected for this question. No action needed.

Sustainability

Questions answered

6/6

Question status

-  High risk: 0
-  Medium risk: 0
-  No improvements identified: 6
-  Not Applicable: 0
-  Unanswered: 0

Pillar notes

-

1. How do you select Regions for your workload?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Choose Region based on both business requirements and sustainability goals

Out of Scope

Notes

SUS_1

Improvement plan

No risk detected for this question. No action needed.

2. How do you align cloud resources to your demand?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Scale workload infrastructure dynamically
Out of Scope
- Align SLAs with sustainability goals
Out of Scope
- Stop the creation and maintenance of unused assets
Out of Scope
- Optimize geographic placement of workloads based on their networking requirements
Out of Scope
- Optimize team member resources for activities performed
Out of Scope
- Implement buffering or throttling to flatten the demand curve
Out of Scope

Notes

SUS_2

2. How do you align cloud resources to your demand?

Improvement plan

No risk detected for this question. No action needed.

3. How do you take advantage of software and architecture patterns to support your sustainability goals?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Optimize software and architecture for asynchronous and scheduled jobs
Out of Scope
- Remove or refactor workload components with low or no use
Out of Scope
- Optimize areas of code that consume the most time or resources
Out of Scope
- Optimize impact on devices and equipment
Out of Scope
- Use software patterns and architectures that best support data access and storage patterns
Out of Scope

Notes

SUS_3

Improvement plan

3. How do you take advantage of software and architecture patterns to support your sustainability goals?

No risk detected for this question. No action needed.

4. How do you take advantage of data management policies and patterns to support your sustainability goals?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Implement a data classification policy
Out of Scope
- Use technologies that support data access and storage patterns
Out of Scope
- Use policies to manage the lifecycle of your datasets
Out of Scope
- Use elasticity and automation to expand block storage or file system
Out of Scope
- Remove unneeded or redundant data
Out of Scope
- Use shared file systems or storage to access common data
Out of Scope
- Minimize data movement across networks
Out of Scope
- Back up data only when difficult to recreate

4. How do you take advantage of data management policies and patterns to support your sustainability goals?

Out of Scope

Notes

SUS_4

Improvement plan

No risk detected for this question. No action needed.

5. How do you select and use cloud hardware and services in your architecture to support your sustainability goals?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Use the minimum amount of hardware to meet your needs
Out of Scope
- Use instance types with the least impact
Out of Scope
- Use managed services
Out of Scope
- Optimize your use of hardware-based compute accelerators
Out of Scope

Notes

SUS_5

Improvement plan

No risk detected for this question. No action needed.

6. How do your organizational processes support your sustainability goals?

✔ No improvements identified

*This question has best practices marked as not applicable by the reviewer

Selected choice(s)

-

Not selected choice(s)

- None of these

Best Practices marked as Not Applicable

- Communicate and cascade your sustainability goals
Out of Scope
- Adopt methods that can rapidly introduce sustainability improvements
Out of Scope
- Keep your workload up-to-date
Out of Scope
- Increase utilization of build environments
Out of Scope
- Use managed device farms for testing
Out of Scope

Notes

SUS_6

Improvement plan

No risk detected for this question. No action needed.